Group Forecast and Furious
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BUSINESS UNDERSTANDING

Financial markets are complex and dynamic systems in which asset prices evolve under the influence of both quantitative factors, such as historical price movements, and qualitative factors, including the arrival of new information and investor sentiments. Traditional approaches to stock market prediction have primarily relied on historical price data and technical indicators; however, these methods often struggle to consistently capture short-term market movements due to the non-linear, volatile nature of financial time series data. Research has shown that financial news and media content can also influence investor behavior and market outcomes by shaping sentiment and expectations.

This project combines these approaches by developing and evaluating a predictive analytics framework that combines structured historical market data with unstructured financial news data processed through NLP techniques. By focusing on next-period return direction classification rather than exact price prediction, the study seeks to assess whether hybrid, multimodal models can enhance predictive performance while remaining aligned with established financial theory and realistic market constraints.

A model that accurately forecasts the price movement of a stock given its price history and sentiment analysis has a large potential monetary value. If an individual is confident that the value of a stock will decrease over the near term, they can sell this stock and invest in a more promising option. Conversely, if the individual is confident that the value of a stock will rise over the near term, they can buy low and receive a profit when they ultimately decide to sell. Each stock can be evaluated individually, and then summed up across a portfolio to receive the total monetary gain.

If we make some initial assumptions including an initial stock investment of $1 MM, a daily decision to buy more of the stock or sell, and a correct decision increases the value of the investment 1% the following day while an incorrect decision will decrease the value of the investment 1% the following day, the estimated value increase of a 52% accurate model over the current state guarantee of a 3.5% annual return (the approximate yield on an annual bond from the U.S. treasury as of January 2026) is approximately $71,033 over the next twelve months. This value would increase with a more accurate model, or if the guaranteed return were to decrease from 3.5%.

DATA ACQUISITION AND PREPARATION

Apple Stock (AAPL): Historical Financial News Data obtained from kaggle.com is the news data which we will be working with for the purpose of this project. This dataset contains financial news data that is related to Apple Inc. (AAPL) and includes information from multiple news articles and their corresponding metadata, sentiment analysis, and relevance to Apple stock. A supplementary dataset contains historical AAPL stock price dataset, which will be combined with the news sentiment dataset to create three possible datasets for the model.

Our target variable will be a binary indicator that represents the directional change in closing price between consecutive trading periods. These will be defined as 1 for an increase and 0 for a decrease or no change. The directional change will be calculated using returns from the previous period. In our case, this includes returns from the previous trading day. Returns are used instead of raw prices because they use a stationary representation that better captures the market's reaction to information such as new sentiment. The returns variable will be constructed using the raw time-series closing price data, and the directional change binary indicator will be constructed using the returns variable. A directional binary indicator was chosen over an exact price prediction for its robustness to noise (classification reduces targets to a simple "up"/"down"), and its relevance to real-life buying and selling stock decisions. This allows us to build classification models that use news sentiment to predict the directional change of the stock price between periods.

The primary predictor variable in this study is the news sentiment score derived from articles related to AAPL. For each time period, news articles are aggregated to produce a sentiment polarity score ranging from −0.99 to 1.00, where negative values indicate negative sentiment, positive values indicate positive sentiment, and values near zero indicate neutral sentiment. In addition to the overall sentiment of the news article, we also include variables that estimate the percentage of positive, neutral, and negative language present. All sentiment dataset variables are lagged up to five days since news articles may impact consumers for an extended period of time. In addition, we include historical price lags, as previous stock price directions may impact future stock price directions.

The strategy of data partitioning between training, validation, and test datasets follows the 60/20/20 rule advised in academic articles (Muraina), as well as other courses of the Applied Analytics program. This rule states that generally, 60% of the entire dataset should be used for training, 20% for model validation, and 20% as a test dataset. This allows the majority of the dataset to be devoted to model training, a sizable portion to evaluate possible models, and another sizable portion to act as a never-before-seen dataset.

A chronological split of the dataset is ideal for this project since it prevents data leakage and look-ahead bias. With this split, future information cannot improve past predictions. The training section of the sentiment dataset encompassed January 2018 through October 2022, resulting in 15,474 rows. The validation set covered November 2022 through October 2023 (7,598 rows) and the test dataset covered November 2023 through October 2024 (6,563 rows). Not every date contained a news article about the AAPL stock, and some dates also contained duplicates, hence the differing number of observations over time periods of the same length.

The AAPL stock price dataset was also partitioned using the same date cutoffs. The resulting datasets had 1,217 rows (training), 251 rows (validation) and 252 rows (test). The sentiment datasets were then aggregated so that each row contained one day and the average sentiment scores in news articles for that day. These dataset was joined on the stock price datasets to achieve cohesive training, validation, and test datasets.

Moving now to the stock price dataset combined with the daily sentiment scores, AAPL stock price was plotted with a simple timeseries. Prices generally increased over time, reaching a peak in January 2022, before decreasing slightly through the end of the year. There were peaks and troughs throughout the entire timeseries, but the most notable increase occurred starting in March 2020. This lines up extremely well with the COVID-19 pandemic, and the stock faltered only slightly when the labor market started to condense at the end of 2022. This is particularly notable since the training dataset has more observations with an “increase” label compared to “decrease” label by an approximate 53/47 margin while the training and validation sets have a more even label distribution.

The AAPL stock price dataset was already fairly clean and free of missing values. The most significant cleaning was done to prepare the data for a classification problem by creating target and predictor variables that represent stock directional movement. The news sentiment dataset required more preprocessing including the combination of title and content columns into one “full text” column and the removal of columns that weren’t relevant to model construction. This helped streamline the dataset, reduce dimensionality, and prevent unnecessary noise from entering the feature space. Finally, the dataset was transformed from article-level observations into daily aggregates. All news articles published on the same date were grouped into a single row representing that day.

The combined dataset encompasses both news article sentiment and AAPL stock price. Each observation contains the daily average sentiment polarity as well as the closing day stock price.
A small number of outliers were identified using the sentiment polarity timeseries. These outliers had abnormally low sentiment polarities, contrasted with positive sentiment scores on the preceding or following days. When directly compared with the stock price timeseries, however, these negative scores

directly coincided with dips in the stock price, especially those in late 2020. These negative sentiment polarities could be very useful to a classification model predicting stock price direction, and as a result, these outliers were elected to stay a part of the dataset.

DATA MODELING AND EVALUATION

The first model in consideration is the logistic regression, a supervised machine learning algorithm used for binary classification. The primary objective is to apply logistic regression to a textual and sentiment dataset in order to classify the directional stock price. Given the broader goal of predicting stock price direction (up or down) using sentiment signals, Logistic Regression serves as an interpretable and computationally efficient baseline model. The logistic regression model was implemented across all three datasets, and tuned using regularization strength and maximum iterations as hyper-parameters.

The second model in consideration is slightly more complex, a support vector machine model. SVMs strive to find the optimal hyperplane that separates the “up” and “down” classes with as minimal error as possible. They are able to use different kernels, which can capture non-linear relationships between variables. SVMs also tend to perform well with NLP datasets, hence the reasoning to use it on the sentiment and combined datasets in particular. After implementing an initial support vector classifier, four hyper-parameters were tuned including kernel, regularization strength, gamma, and maximum iterations.

The third model implemented is a FinBERT model, which was implemented solely on the news sentiment dataset. The FinBERT-based approach focused on enhancing text representation by generating finance-specific sentiment features from the full_text column and combining these with the existing sentiment variables to predict next-period stock price direction. Rather than relying on TF-IDF representations, FinBERT was used to extract structured sentiment probabilities (positive, negative, and neutral), along with derived features such as polarity, which were then used as inputs into a logistic regression classifier. This approach leverages domain-specific language understanding, allowing the model to better capture financial tone and context embedded in news text while maintaining a lightweight and CPU-friendly modeling pipeline. During fine-tuning, hyperparameters included regularization type and regularization strength.

Our final model, XGBoost, is a complex model with a tree-based algorithm. This model builds on the basic decision tree by implementing boosting, a method that uses each prior decision tree to inform the next. The model also uses gradient descent, which attempts to minimize the loss in the model’s performance over the iterations. This gives it similar attractive properties as a neural network, but it requires less data to train on, which is attractive for our project since we only have approximately two thousand observations in the combined dataset. The model was implemented on both the historical price and combined datasets. Optimized hyper-parameters include the number of estimators, maximum depth, and learning rate.

The primary metric used for best model consideration is accuracy with F1 macro average in support. Accuracy is the primary metric because a correct prediction in the increase or decrease of stock price is equally valuable. In the initial assumptions of our product, we assumed that the client made equal profit through investing (associated with an increase in the AAPL stock price) or divesting (associated with a decrease in the AAPL stock price). Therefore, we care most about the percent of correct overall predictions, followed by a metric that balances performance for each stock direction (f1 score). The misclassification error can also be viewed across models in the chart below.

The following table shows the results of each model-dataset combination:

	Training Misclassification Error (1-Accuracy)	Validation Misclassification Error (1 - Accuracy)	Training F1 (Macro Average)	Validation F1 (Macro Average
Price Logistic	0.4856	0.4582	0.4301	0.4512
Sentiment Logistic	0.4717	0.4701	0.3457	0.3464
Combined Logistic	0.4678	0.4878	0.4740	0.4577
Price SVM	0.4848	0.5259	0.5152	0.4740
Sentiment SVM	0.4717	0.4702	0.3457	0.3464
Combined SVM	0.4521	0.4472	0.4409	0.4359
Price XGBoost	0.4569	0.5458	0.5367	0.4463
Combined XGBoost	0.3919	0.4268	0.5225	0.4590
Sentiment FINBERT	0.4552	0.4582	0.4174	0.4594

The XGBoost model on the combined dataset had the highest accuracy and lowest misclassification error on the validation dataset. It is not surprising that a model using the combined dataset had the highest accuracy, this model has more information available than any model using the sentiment or historical price datasets. In addition, the XGBoost model is the most complex model of the three using the combined dataset, and the sole model using decision tree methodology, which was improved even further with boosting. This model also likely succeeded because it was able to learn quickly on a small dataset of approximately two thousand observations, where other complex models (like FINBERT) took longer to understand the patterns between features.

In general, the more complex models tended to have lower errors on the validation datasets. In addition, the models using only historical price tended to perform the worst followed by sentiment data. This is likely due to the complex nature of the dataset needing complex models to understand patterns, and the sentiment and historical price datasets lacking critical information when used individually.

However, when evaluated against the test dataset, the model underperformed, only reaching 52.22% accuracy. This model decreased substantially in accuracy between the training, validation, and test datasets (as shown below). This shows signals of high variance and is likely due to the model overfitting on the training dataset, which scaled poorly as the datasets changed.

Model	Training Accuracy	Validation Accuracy	Testing Accuracy
Combined XGBoost	60.81%	57.32%	52.22%

IMPROVING DATA QUALITY

With test dataset performance being underwhelming, we took steps to improve the quality of inputs in the training dataset. Firstly we created lagged versions of the raw AAPL share price, which means it captures the actual numerical value of the stock at previous time steps. Specifically, for each lag from 1 to 5, the .shift(i) function moves the price data down by i rows, so that each new column represents the stock price from prior days .As a result, the model can learn patterns related to magnitude, volatility, and trends over time where previously it only had access to previous stock price direction. These features are particularly useful for more complex models that benefit from richer numerical input, as they preserve the full information contained in the original price series.

The second step is a technique called SMOTE, which creates synthetic observations for the minority class. This is particularly important for this dataset because the “increase” stock price observations outweighed the “decrease” observations by 53% to 47%. This allowed the model to create a baseline accuracy of 53% simply by predicting all observations as “increase”. SMOTE evens out the discrepancy between the classes, and reduces the possibility of a model optimizing accuracy by predicting a single class.

We also used the bootstrapping method to see if that would have a positive impact on the model's performance. Bootstrapping is a resampling technique that creates new training datasets by randomly sampling the original dataset with replacement. This means that the same data points can appear multiple times within a sample. Some data points may also not be included at all. Bootstrapping is often used when a dataset is small and more observations would be helpful for modeling. In our case, we are working with daily stock price data for a single company. Since there is a finite number of data points for our case (stock data only exists for as long as a company has been public) bootstrapping is a logical way to modify our data.

With all three techniques implemented, the validation accuracy increased to a promising 70%. This is much higher than the 57% validation accuracy achieved with the previous model. The macro average for the mode with improved data is also much higher than the original model. This is likely due to the increase of sample size (SMOTE and bootstrapping both helped with this) compared to the previous training dataset, which was severely lacking observation count. The conversion of historical price lags to quantitative variables also provided the model with additional information, which helped inform more accurate predictions.

	Accuracy		Macro Average F1 Score	
	Validation Dataset	Test Dataset	Validation Dataset	Test Dataset
Original XGBoost	57%	52%	0.46	0.39
XGBoost with SMOTE and Boostrapping	71%	51%	0.71	0.45

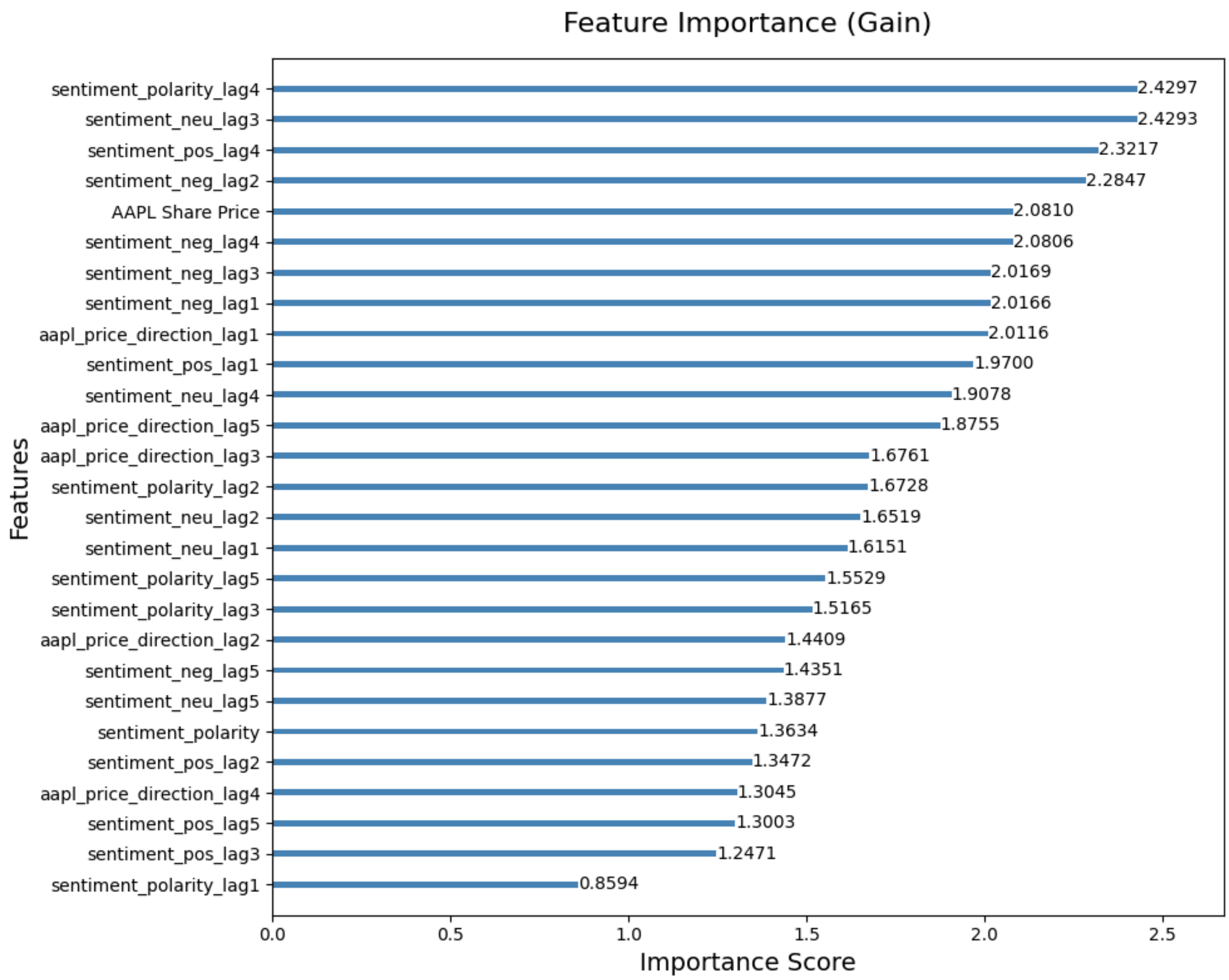
When implemented on the test dataset, accuracy was slightly lower than the previous week’s model, ticking down to 51%. This suggests the model is still heavily overfitting on the training and validation datasets, and is unable to extend similar performance to the test dataset.

Our secondary performance metric, the macro average F1 score, did increase from 0.39 to 0.45 in week 10 after the implemented data techniques. The performance across the two classes was much more evenly distributed, with the lowest class f1 score increasing from 0.11 to 0.27. Interestingly, the model implemented this week did a much better job implementing the "decrease" class over the "increase" class, a change from previous weeks' models. This suggests the SMOTE technique did indeed work, but it now caused the model to be slightly too cautious in predicting increases in stock price direction.

Next, we attempted to remove some bias from the model. There are many potential sources of bias including a lack of diversity within news sources and the skew towards negativity. However, one that we attempted to address is the reality of Apple being reported on not just because of their own company news, but because of their business competition or other companies in the writer's portfolio. This is especially evident when Apple is being compared to other large technology companies, such as Amazon or Tesla. The news article will comment on "large tech companies" stock performance, and will tag Apple in the keywords, but it doesn't actually refer to the individual company when speaking in generalities. In other words, Apple's news sentiment is biased by group association. This could systematically inflate (or deflate) Apple's stock price association, even on days that Apple stock is not being directly reported on. We account for this by retraining a model that only explicitly mentions the Apple stock in title or content.

	Accuracy		Macro Average F1 Score	
	Validation Dataset	Test Dataset	Validation Dataset	Test Dataset
XGBoost with SMOTE and Boostrapping	71%	51%	0.71	0.45
XGBoost with Reduced Bias	57%	55%	0.57	0.55

The XGBoost model with reduced bias actually surpassed the XGBoost model with SMOTE and bootstrapping on the test dataset in both accuracy and F1 score, despite achieving lower performance on the validation set. This suggests that the bias removal helped with overfitting, which was a large problem in previous weeks, and removed some of the "noise" that previously caused this model to train on false patterns in the data. In addition, the F1 scores for the "increase" and "decrease" classes were identical, a balance no prior model had been able to achieve. This model is selected for deployment.



In addition to model performance, we also investigated the features that contributed most to model performance (decreases in the loss function). Overall, the news sentiment scores ranked much higher than the historical price direction lags. This is perhaps not too surprising since the models trained on solely news sentiment data performed much better than the models trained solely on the historical price data in terms of accuracy and F1 scores. Of the historical price lags, the most recent direction lag was ranked highest, showing that recent price data is still most effective in predicting next-day price direction.

Within the sentiment data, the 2 and 3 day lags between the negative, neutral, and polarity scores were all ranked high in importance. This information suggests that it takes a couple days for news articles to impact stock price direction, and articles containing an extreme amount of negative language as well as extreme overall sentiment (measured through polarity) are correlated with a consumer's decision to buy or sell stock. Noticeably absent is the percentage of positive language, this ranked very low in feature importance.

DEPLOYMENT

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This model will be deployed in batch mode on a daily schedule over real-time inference. Since the model uses data aggregated on a daily level, the model does not need to make an inference more than once a day. Batch mode deployed on a daily schedule is a perfect candidate because it will generate predictions based on the previous day's closing stock price and news sentiment and deliver to the client on a daily cadence. It can do this without racking up high costs like a real-time inference deployment would, and it uses far fewer computational resources.

A business metric that should be tracked in production is revenue generated. We made an initial assumption of a $1MM investment, and correct predictions of this model could allow the client to grow that investment by buying stock at relatively low prices and selling stock at relatively high prices. Generating substantial revenue could justify the use of a machine learning model over headcount for a human day trader, and allow the business to grow while keeping expenses limited.
Upon deployment, there are a few potential risks that would be important to keep in mind.

We will primarily monitor test dataset accuracy and macro average F1 score in production. If our model is within the green threshold (at or better than 0.53), it will remain in production with our standard monitoring and routine retraining. If it falls within the yellow threshold (0.51 - 0.52), the model will also remain in production but will require more monitoring. We would also have an analyst examine possible data or concept drift. If the model falls into the red threshold (< 0.50) it would be pulled from production. Similar to how we built our initial model, we could use this time to rebuild a new model using what we know about data or concept drift.

Our model would need to be trained regularly instead of kept static. Both stock prices and news sentiment are constantly changing. We would need to update our model with new data as frequently as possible in order to avoid concept drift, when historical patterns become less important in the future. Since companies like Apple report their earnings every quarter, it is logical to retrain our model each quarter with updated stock price and new sentiment data. This will also combat data drift, which may happen if the distribution of predictor variables changes in relation to the training data